

PARTIAL DIFFERENTIAL EQUATIONS

Arshad Afzal, PhD

Partial Differential Equations - Introduction

Partial differential equation (PDE)

- A partial differential equation (PDE) is an equation involving one or more partial derivatives of an (unknown) function, u , that depends on two or more variables, often time t and one or several variables in space.
- The order of the highest derivative is called the order of the PDE.
- A PDE is linear if it is of the first degree in the unknown function u and its partial derivatives, otherwise it is nonlinear.
- A linear PDE is called homogeneous if each of its terms contains either u or one of its partial derivatives, otherwise it is nonhomogeneous.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

One - dimensional wave equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

One - dimensional diffusion equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Two - dimensional Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

Two - dimensional Poisson equation

- Example of linear, but variable coefficient PDE is:

$$a \frac{\partial u}{\partial t} + bx \frac{\partial u}{\partial x} = 0$$

- Examples of non – linear PDEs are:

$$u \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y} = 0$$

$$a \left(\frac{\partial u}{\partial x} \right)^2 + b \frac{\partial u}{\partial y} = 0$$

Partial Differential Equations - Introduction

Concept of solution

- A solution of a PDE in some region R of the space of the independent variables is a function that has all the partial derivatives appearing in the PDE in some domain D containing R , and satisfies the PDE everywhere in R .
- If u_1 and u_2 are solutions of a homogeneous linear PDE in some region, R , then

$$u = c_1 u_1 + c_2 u_2$$

with any constants c_1, c_2 is also a solution of that PDE in region, R

- Question: Find solution of below PDE

$$u_{xx} - u = 0$$

One-Dimensional Wave Equation

1-D wave equation

- Model of a vibrating elastic string

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$u(0, t) = 0, \quad u(L, t) = 0, \quad \text{for all } t \geq 0$$

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x), \quad (0 \leq x \leq L)$$

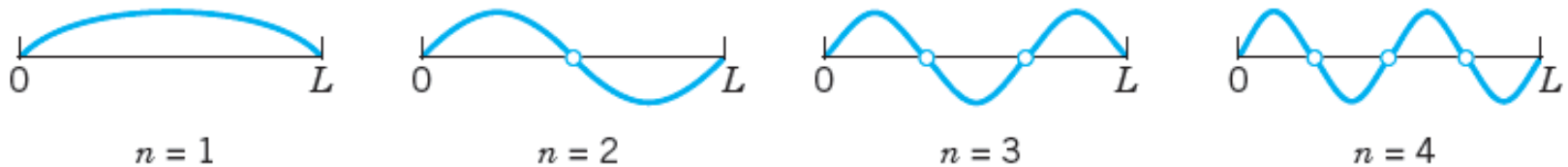
Solution Procedure

- Method of separating variables, or product method

$$u(x, t) = F(x)G(t)$$

$$u_n(x, t) = (B_n \cos \lambda_n t + B_n^* \sin \lambda_n t) \sin \frac{n\pi}{L} x, \quad n = 1, 2, \dots$$

- These functions are called the eigenfunctions, or characteristic functions, and the values $\lambda_n = cn\pi/L$ are called the eigenvalues, or characteristic values, of the vibrating string. The set $\{\lambda_1, \lambda_2, \dots\}$ is called the spectrum.



Normal modes of vibrating string

One-Dimensional Wave Equation

Solution: Fourier series

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} (B_n \cos \lambda_n t + B_n^* \sin \lambda_n t) \sin \frac{n\pi}{L} x$$

One-Dimensional Wave Equation

Special case

- Initial velocity, $g(x) = 0$

One-Dimensional Wave Equation

D'Alembert's solution of the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$u(0, t) = 0, \quad u(L, t) = 0, \quad \text{for all } t \geq 0$$

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x), \quad (0 \leq x \leq L)$$

Solution Procedure

- Change of variables

$$v = x + ct, \quad w = x - ct$$


$$u(x, t) = \Phi(x + ct) + \Psi(x - ct)$$

- The above solution is known as D'Alembert's solution of the 1-D wave equation

One-Dimensional Wave Equation

D'Alembert's solution satisfying the initial condition

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x)$$



$$u(x, t) = \frac{1}{2} [f(x + ct) + f(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$



Types of PDEs

- Consider a general form of PDEs

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} = F(x, y, u, u_x, u_y)$$

The above equation is called quasi – linear because it is linear in the highest derivatives.

- Based on the above equation, there are three types of PDEs depending on a parameter, Δ defined below

$$\Delta = B^2 - AC$$

$$\Delta = B^2 - AC > 0 \text{ (Hyperbolic equation)}$$

$$\Delta = B^2 - AC = 0 \text{ (Parabolic equation)}$$

$$\Delta = B^2 - AC < 0 \text{ (Elliptic equation)}$$

Characteristic equation

$$Ay'^2 - 2By' + C = 0, \quad y' = \frac{dy}{dx}$$

The solution of the above equation are called characteristics of the PDE, and are paths in solution domain along which information propagates.

Find the solution of the wave equation satisfying the following conditions

$$u(0, t) = 0, \quad u(L, t) = 0, \quad \text{for all } t \geq 0$$

$$u(x, 0) = f(x), \quad u_t(x, 0) = 0, \quad (0 \leq x \leq L)$$

where

$$f(x) = \begin{cases} \frac{2k}{L} x, & 0 < x < \frac{L}{2} \\ \frac{2k}{L} (L - x) & \frac{L}{2} < x < L \end{cases}$$

